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Effects of a low- or high-frequency colostrum feeding protocol on immunoglobulin G absorption in newborn calves

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ABSTRACT

Calves might experience an upper limit of IgG absorption from colostrum ingestion at birth, but it is not clear whether the total IgG mass fed in the first meal or feeding frequencies can saturate the IgG transport mechanism and therefore limit IgG absorption. The objective of this study was to determine whether different colostrum replacer (CR) feeding frequencies affect serum IgG levels or apparent efficiency of absorption (AEA) in neonatal calves. Male Holstein calves (n = 40) were separated from their dams immediately after parturition and randomly assigned to receive CR [12% of birth body weight (BW)], following either (1) a low-frequency (LF; n = 20) or (2) a high-frequency (HF; n = 20) feeding protocol. Low-frequency calves received 2 CR meals (8% and 4% birth BW within 1 h after birth and 12 h after first CR feeding, respectively), whereas HF calves received 3 CR meals (4% of BW for each meal; within 1 h after birth, 6, and 12 h after first CR feeding). The CR powder fed had a dry matter IgG concentration of 30% and an IgG concentration of 70.5 g/L when reconstituted. All CR was fed via esophageal tube within 1 h after birth. Calves were bottle-fed pasteurized milk (5% birth BW) at 24, 36, and 48 h after the first CR feeding. Blood was collected before first CR feeding and at the following intervals post-CR feeding: every 2 h until 18 h; every 3 h from 18 to 30 h; and every 6 h from 30 to 48 h after the first CR feeding. Serum IgG values at 24 h did not differ between LF and HF (25.79 ± 0.93 and 25.66 ± 0.88 g/L, respectively). In the first meal, calves fed LF ingested a higher total IgG mass than HF (257.98 ± 4.16 g and 126.72 ± 4.05 g, respectively); however, AEA at 24 h did not differ for calves fed HF or LF (27.68 ± 1.16% and 27.63 ± 1.26%, respectively). The IgG area under the curve (AUC) at 24 h was greater for calves fed LF than HF (443.13 ± 15.17 and 379.59 ± 13.99 g of IgG/L × h, respectively). Additionally, AUC at 6 h, 12 h, and 48 h were greater for calves fed LF than HF. These results indicate that, although LF calves had a greater AUC, HF calves were still able to absorb IgG in the second and third meal, allowing HF calves to achieve serum IgG levels similar to those of LF calves at 24 h. In addition, the provision of 3 meals at 70.5 g/L of IgG within the first 12 h of life did not result in added benefits to serum IgG or AEA levels.
Key words: calf, colostrum replacer, IgG absorption

INTRODUCTION

Newborn calves depend on colostrum ingestion to acquire circulating IgG during their first days of life (Lombard et al., 2020). Successful transfer of passive immunity is achieved when calves reach serum IgG concentrations above 10 g/L; in contrast, concentrations <10 g/L are considered as failed transfer of passive immunity (Quigley, 2004; Shivley et al., 2018). To achieve this threshold, it is recommended that calves consume 10% of their birth BW of colostrum at birth (3 to 4 L; Besser et al., 1991; Godden et al., 2019) to ingest at least 150 to 200 g of IgG (Quigley et al., 2002) within 2 h after birth and achieve transfer of passive immunity. Nevertheless, early studies suggest that calves need to ingest 300 to 400 g of Ig to achieve protection against entero-pathogens (Roy, 1980). This mass of Ig should be fed within 2 h after birth, as delayed feeding negatively affects IgG absorption (Fischer et al., 2018), but guidelines for the number of colostrum meals and volume of colostrum fed per meal have not been clearly established. Although transfer of passive immunity is achieved at a serum IgG concentration of 10g/L, recent recommendations state that calves should achieve a much higher serum IgG concentration (Lombard et al., 2020). As a result, higher colostral IgG ingestion than current recommendations would need to be delivered in the first colostrum meal.

Although it is known that delaying colostrum feeding by 6 and 12 h reduced IgG absorption by 26% compared with calves fed immediately after birth (Fischer

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et al., 2018), it is not known how the different number of meals affects transfer of passive immunity or IgG absorption. Notwithstanding, it has been reported that in the United States, small farms (30 to 99 cows) feed ~3.79 L in the first 24 h, and large farms (≥500 cows) feed ~5.67 L or more (NAHMS, 2014). However, this data does not specify the number of meals fed. Jaster (2005) evaluated the effects of feeding 1 versus 2 colostrum meals within 12 h to Jerseys calves. The author reported that feeding 2 L of high-quality colostrum (84 g IgG/L) in 2 meals (0 h and 12 h) instead of 1 meal (4 L at 0 h) resulted in higher apparent efficiency of absorption (AEA) and serum IgG levels at 24 h. In contrast, other reports state that Holstein calves fed 1 meal of 4 L (60.1 g IgG/L), followed by a 2-L meal at 12 h, resulted in higher serum IgG levels than 2 meals of 2 L (Morin et al., 1997). In addition, Abuelo et al. (2021) compared the effects of 1 versus 2 meals of colostrum, and, although possible saturation level was not the focus of the study, it was reported that calves fed a second colostrum meal at 5 to 6 h had lower odds of failed transfer of passive immunity than calves fed just 1 meal at birth. Given these findings, mulitple colostrum meals may aid in improving successful transfer of passive immunity; however, the effects of multiple meals on absorption of IgG are not well understood.

Total IgG mass is directly related to IgG concentration and increases as greater volumes of colostrum are fed, but AEA simultaneously decreases and is negatively correlated with the amount of IgG fed (Besser et al., 1985; Saldana et al., 2019). This decrease in AEA, when high total masses of IgG are fed at birth, is thought to occur due to the physiological limitation of the intestine to absorb IgG within a determined time (Besser et al., 1985). This phenomenon has also been defined as "gut saturation" or achieving an upper limit of IgG absorption (Saldana et al., 2019). Besser et al. (1985) proposed 2 possible hypotheses for this upper limit of absorption when high amounts of IgG are fed. First, that shared macromolecular transport mechanisms of IgG are saturated, or, second, that the calf regulates its serum IgG concentrations when a threshold level is met (Besser et al., 1985). In support of the competitive binding hypothesis, Davenport et al. (2000) reported that adding casein to colostrum supplements or maternal colostrum reduced IgG absorption, and Lopez et al. (2020) found that the removal of casein in colostrum replacer (CR) increased AEA. Davenport et al. (2000) discussed that intake of further proteins or casein above a limited macromolecular transport would not be absorbed, and, as a result, circulating IgG in the blood could decrease. In consequence, AEA can be compromised.

Thus, we hypothesized that reducing total IgG mass fed at birth by increasing the number of colostrum meals would increase AEA. Thus, the objective of this study was to evaluate the effects of 2 different CR feeding frequencies, low and high, by delivering a determined volume according to birth BW in 2 or 3 meals. Also, we aimed to evaluate whether an upper limit of IgG absorption exists or whether AEA is decreased when a larger meal is fed at birth.

MATERIALS AND METHODS

The animal use protocol was approved by the Animal Care Committee of the University of Guelph (Guelph, ON, Canada; AUP no. 4488). This study was conducted at a commercial dairy farm with a total herd of ~3,500 animals and ~1,600 milking cows in Aylmer, Ontario, Canada. This farm was chosen due to their willingness to participate in the research project and the availability of a high frequency of calvings over a short period, which would reduce any calf-to-calf variation that may have occurred in different seasons.

Calving and Neonatal Calf Management

Primiparous and multiparous Holstein cows were separated from the close-up pen and moved to a maternity pen approximately 1 to 2 d before parturition. Maternity pens were bedded daily with shavings and sanitized between calvings. Cows were monitored during the calving process, and assistance was provided when needed. Male Holstein calves born in the maternity pen were immediately separated from their dams to prevent colostrum suckling. Afterward, calves were thoroughly dried with 3 clean towels, the navel was dipped with iodine, and a navel clamp was placed on the distal aspect of the navel. Calves were weighed with a digital scale (Brecknell Scales, PS1000, Avery Weigh-Tronix, LLC) and moved to individual calf hutches bedded with shavings (Starter Pen, AGRI-6000B, Agri-Plastics). In between calves, pens were cleaned and disinfected with alcohol (70%) and a chlorhexidine solution (Germi-Stat 4%, Germiphene Corporation). Sanitized pens were allowed to dry for 1 d before a new calf was assigned.

Animal Experiment, Colostrum Replacer, and Milk Feeding

Male Holstein calves (n = 40; 20 per treatment) born from November 2020 to December 2020 were enrolled in this experiment. All newborn calves were randomly assigned to treatment to receive CR following either (1) a low-frequency (LF) or (2) a high-frequency (HF) feeding protocol. Calves in the LF treatment were offered CR within 1 h after birth (8% of birth BW) and 12 h (4% of birth BW) after the first CR feeding. Meanwhile, the HF calves received CR within 1 h after birth, and 6 and 12 h after the first CR feeding (4% of birth BW per meal). Regardless of feeding frequency treatment, all calves were fed CR only at a total of 12% of birth BW. The randomization sequence was generated in Microsoft Excel (version 16.16.21, Microsoft Corporation) using the RAND function per block. The CR powder used was derived from bovine dried colostrum, contained IgG at 30% DM, and was reconstituted with water to 70.5 g of IgG/L when fed (Saskatoon Colostrum Company Ltd.). The composition analysis of the CR is presented in Table 1. In addition, the Quality Assurance Laboratory from Saskatoon Colostrum Company Ltd. (Saskatoon, SK, Canada) reported that coliforms, *Salmonella* spp., and *Escherichia coli* were absent. Reconstitution of CR was performed as follows: (1) weighing CR powder, (2) weighing warm (40°C) water, (3) reconstituting the solution by adding CR powder to a bowl and slowly adding water while mixing with a whisk for approximately 2 min, and (4) weighing the reconstituted solution to verify it matched with the final mass to be fed. After verification, reconstituted CR was transferred to a 4-L esophageal tubing bottle. All CR meals were offered via an esophageal tube, and calves were not allowed to voluntarily consume colostrum. The first CR feeding was fed to all calves 1 h after birth. After the corresponding CR meals, all calves were fed (5% of BW) pasteurized waste milk at 24, 36, and 48 h after the first CR feeding via nipple bottle. Milk was heated to ~39°C in a warm water bath before feeding. Refused milk was weighed and recorded for every feeding. Calves were weighed at 48 h immediately after the last milk feeding when their experimental enrollment concluded.

Blood Sampling

At approximately 10 to 20 min after birth, a jugular catheter was placed, as described by Fischer et al. (2018), to enable frequent blood sampling for 48 h. Before catheter placement, one side of the calf's neck was shaved with clippers (Wahl Basic Pet Clipper Kit, model no. 58108), scrubbed with a chlorhexidine solution (Germi-Stat 4%, Germiphene Corporation), and sprayed with isopropanol (70%). Thereafter, a 2-inch, 16-gauge catheter (Thermo Fisher Scientific) was placed in the jugular vein. As soon as the catheter was placed, a baseline blood sample was taken (0 h). Blood samples (7 mL per collection) were then collected at the following intervals: every 2 h until 18 h; every 3 h from 18 to 30 h; and every 6 h from 30 to 48 h after the first CR feeding. All blood samples that coincided with a CR or milk meal were taken before feeding. Blood samples (7 mL) were transferred to a serum blood collection tube (Vacutainer; Becton, Dickinson and Co.). Immediately after every collection, 7 mL of saline and 1 mL of heparinized saline were flushed back into the catheter. Samples in serum collection tubes were allowed to clot at room temperature for approximately 1 to 2 h before centrifugation. Samples were centrifuged at 3,000 × *g* at 4°C for 20 min. Serum collected after centrifugation was transferred into three 1.5-mL microcentrifuge tubes in equal aliquots and frozen at −20°C until further analysis.

Table 1. Composition analysis of colostrum replacer used to feed calves assigned to either low or high feeding frequency treaments[1]

Test description	Specification	Result
IgG, % (minimum)	30	30
Crude protein, % (minimum)	50	60.7
Crude fat, % (minimum)	14	14.3
Lactose, %	N/A	8.0
Moisture, % (maximum)	7.0	6.4

[1]Composition analyses were performed by the Quality Assurance Laboratory of the Saskatoon Colostrum Company Ltd. (Saskatoon, SK, Canada). N/A = not applicable.

IgG and Serum Total Protein Analyses

Serum was thawed and re-centrifuged at 3,000 × *g* for 20 min at 4°C to separate the supernatant, which was then transferred to a new 1.5-mL microcentrifuge tube. Samples were then refrozen at −20°C and shipped overnight on ice to the Saskatoon Colostrum Company Ltd. Quality Assurance Laboratory (Saskatoon, SK, Canada) for IgG and serum total protein analyses. Serum IgG was determined by radial immunodiffusion, as described by Chelack et al. (1993), with modifications according to Shivley et al. (2018). Serum IgG values were determined for all time points and used to represent IgG dynamics during the first 48 h of life. The AEA formula used was adapted from Quigley and Drewry (1998), Quigley et al. (2002), and Saldana et al. (2019), where

$$\text{AEA} = \{[\text{birth weight (kg)} \times 0.09 \times \text{serum IgG (g/L at 24 h)}]/\text{total IgG fed (g)}\} \times 100.$$

Additional parameters calculated with serum IgG data included the following: time to reach maximum concentration, and maximum concentration reached. Also, the positive incremental area under the curve (AUC) was calculated using the trapezoidal rule (Cardoso et al., 2011) at 6 h (AUC_6), 12 h (AUC_{12}), 24 h (AUC_{24}), and 48 h (AUC_{48}) after the first CR feeding.

Table 2. Parameters (mean ± SEM) in newborn calves fed low-frequency (LF; n = 19) or high-frequency (HF; n = 20) colostrum replacer (CR) within the first 12 h after the first CR feeding

Item[1]	Treatment[2] LF	HF	SEM	Contrast *P*-value
BW at birth, kg	46.5	45.6	0.96	0.50
BW at 48 h, kg	48.1	47.3	0.95	0.53
Baseline serum IgG, g/L	0.8	0.5	1.27	0.78
IgG_{24}, g/L	25.8	25.7	1.28	0.92
STP_{24}, g/dL	6.3	6.3	0.26	0.92
AEA, %	27.6	27.7	1.26	0.97
FPTI, %	0	0	—	—
IgG C_{max}, g/L	29.8	27.7	0.91	0.11
IgG T_{max}, h	21.0	22.0	0.94	0.43
AUC_6	36.5	25.1	2.60	0.003
AUC_{12}	145.2	105.9	5.91	<0.01
AUC_{24}	443.1	379.6	15.17	0.004
AUC_{48}	997.8	866.8	33.5	0.008

[1]IgG_{24} = serum IgG concentrations at 24 h after colostrum feeding; STP_{24} = serum total protein concentrations at 24 h after colostrum feeding; AEA = apparent efficiency of absorption, calculated as {[birth weight (kg) × 0.09 × serum IgG (g/L 24 h)]/total IgG fed (g)} × 100; IgG C_{max} = maximum concentration; IgG T_{max} = time to reach maximum concentration; AUC_6, AUC_{12}, AUC_{24}, and AUC_{48} = incremental area under the curve during the first 6, 12, 24, and 48 h after colostrum feeding, respectively; FPTI = failure of passive immunity (serum IgG_{24} <10 g/L).

[2]LF = colostrum replacer fed (total of 12% of birth BW) in 2 meals (8 and 4% of birth BW; within 1 h after birth and 12 h after first CR feeding, respectively); HF = colostrum replacer fed (total of 12% of BW) in 3 meals (4% of BW; within 1 h after birth, and 6 and 12 h after first CR feeding).

Sample Size

An anticipated mean serum IgG concentration was selected from Fischer et al. (2018) due to the similarity of the colostrum feeding methodology. Fischer et al. (2018) reported an average serum IgG value of 22.30 g/L with a standard error of the mean of 1.40 g/L. The standard deviation (SD) was determined to be 4.20 g/L, which was used to calculate a coefficient of variation of 18.83%. Using this value, we calculated the total replicates needed per treatment group using an 80% power and α of 0.05, as outlined by Berndtson (1991). To detect a difference of 20% in serum IgG values compared with the control, 19 animals were needed per group, for a total sample size of 38 calves. To account for 5% mortality, it was then determined that 20 animals per treatment were needed, for a a total of 40 newborn calves.

Statistical Analysis

Data were analyzed using SAS (University Edition, SAS Institute Inc.). A total of 40 male Holstein calves were included in the analysis (n = 20 per treatment), but one calf was removed due to an error when reconstituting CR powder and consequently total IgG consumed. The final data set therefore included 19 calves fed LF and 20 calves fed HF. Residual distribution for each variable was assessed for normality (using Shapiro-Wilk as a criterion) and homoscedasticity using the UNIVARIATE and PLOT procedures. Statistical significance was declared at $P \leq 0.05$. All values are reported as mean ± SD. Effects of colostrum treatment on serum IgG values collected over time were analyzed as a repeated measures design using the GLIMMIX procedure. The model contained the fixed effects of treatment, time, and their interaction, and the random effect of calf. Time (hour) was specified as the repeated measure. Non-repeated measurements were analyzed using the MIXED procedure. The model included the fixed effect of treatment, time, and an interaction of treatment by time, and the random effect of calf. Least squares means are presented with the respective standard errors. Differences between least squares means were compared using the SLICE option, and contrast statements were performed for birth BW, BW at 48 h, serum total protein concentrations at 24 h after colostrum feeding, serum IgG concentrations at 24 h after colostrum feeding, AEA, and baseline IgG.

RESULTS

Body Weight, Colostrum, and Milk Feeding

Birth BW for LF and HF calves did not differ (P = 0.50; Table 2). Additionally, BW was not different (P = 0.53) between treatments at 48 h after colostrum and milk feedings. Calves fed LF were fed a total IgG mass of 257.9 and 128.9 g in the first (0 h) and second (12 h) CR meals, respectively, whereas HF calves were fed 126.7, 126.7, and 126.7 g at the first (0 h), second (6 h), and third (12 h) CR meals, respectively. Overall, LF and HF were fed a total IgG mass of 386.9 and 380.1 ± 8.1 g (P = 0.53), respectively. A milk consumption effect was found for all calves (P < 0.001), where calves drank more milk in the second (36 h) and third (48 h) milk feedings than in the first milk feeding (24 h). Although LF calves were fed a higher amount of colostrum at birth (8% of BW) than HF calves (4% of BW), milk consumption in the first milk feeding did not differ (P = 0.96).

Effect of 2 Feeding Frequencies on IgG Absorption

Baseline serum IgG concentrations at birth were negligible and not different (P = 0.78; Table 2) between LF and HF calves. Feeding colostrum twice versus 3 times did not affect serum IgG values at 24 h (P = 0.92; Figure 1; Table 2). According to serum IgG categories specified by Lombard et al. (2020), 55.56 and 44.44% of LF calves were in the good (18.00 to 24.90 g/L) and excellent (≥25.00 g/L) categories, respectively, whereas 38.10% and 61.90% of HF calves were in the good and excellent categories, respectively. The percentages of calves in either excellent or good category were not different between LF or HF calves ($P > 0.05$). None of the calves fed either LF or HF were classified as having poor (<10.00 g/L) or fair (10.00 to 17.90 g/L) transfer of passive immunity. Additionally, serum total protein values at 24 h were not affected ($P = 0.92$; Table 2) by feeding LF or HF colostrum treatments (Table 2). We found a treatment by time effect ($P = 0.003$) on serum IgG concentrations throughout the experimental period (Figure 1). However, simple sliced effects demonstrated mean differences only in serum IgG between treatments at 8 h ($P < 0.01$; Figure 1).

Apparent efficiency of absorption at 24 h did not differ ($P = 0.97$) between LF and HF calves (Table 2). In addition, calves fed either LF or HF did not have a different maximum concentration ($P = 0.11$; Table 2). Similarly, time to reach maximum concentration for LF and HF did not differ ($P = 0.44$; Table 2). Calves fed LF had a higher ($P < 0.01$) AUC_6 than HF calves (Table 2). Similarly, LF calves had a higher ($P < 0.01$) AUC_{12} than HF calves (Table 2). The same scenario was observed for AUC_{24}, where LF calves had a higher value ($P < 0.01$) than HF calves (Table 2). In addition, LF calves had a higher AUC_{48} ($P < 0.01$) than HF calves (Table 2). Overall, it seems that HF calves were able to absorb enough IgG from meals offered at 6 and 12 h and equalize serum IgG concentrations at 24 h compared with calves fed LF, even though it is known that IgG absorption decreases as time after birth increases (Stott et al., 1979a).

DISCUSSION

We hypothesized that increasing CR feeding frequency from 2 to 3 meals would increase serum IgG concentrations at 24 h and AEA. This possible improvement in IgG absorption could be related to a more efficient IgG absorption when feeding lower masses of IgG compared with higher IgG masses in a certain period of time that could surpass an upper limit of absorption (Besser et al., 1985; Saldana et al., 2019). Nevertheless, results from this study show that serum IgG and AEA levels were not affected by feeding newborn calves either 2 or 3 CR meals postnatally. This might indicate that 2 or 3 colostrum meals already reduce the upper limit of IgG absorption a calf might experience compared with when they are fed one large meal at birth.

Apparent efficiency of absorption of colostral IgG seems to decrease when large colostrum meals or high masses of IgG are fed at birth (Saldana et al., 2019), but this was not seen in the current study. It has to

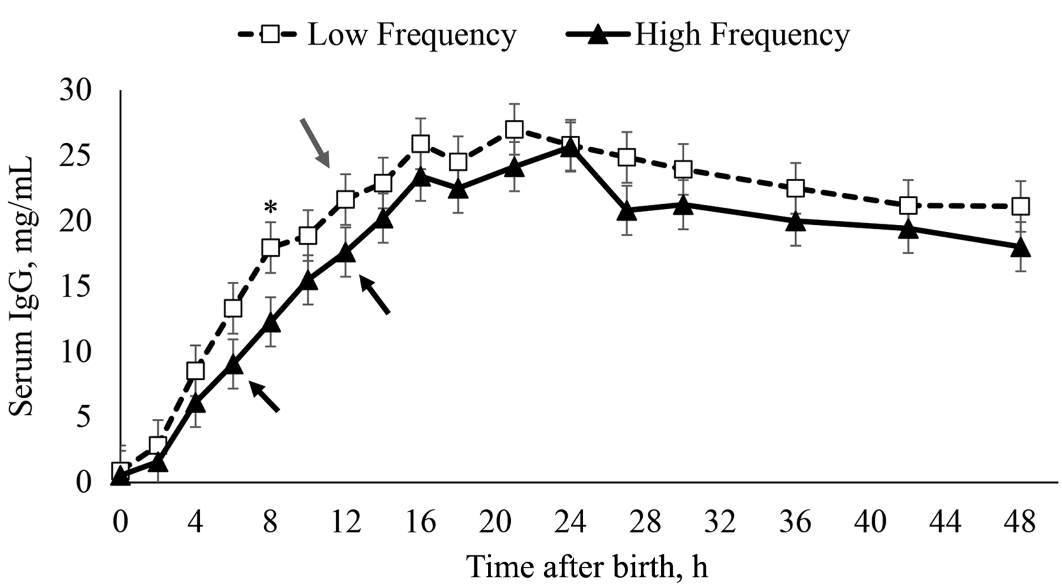

Figure 1. Serum IgG concentrations (mean ± SEM) for calves fed at low frequency (LF; within 1 h after birth and 12 h after first CR feeding) or high frequency (HF; within 1 h after birth, and 6 and 12 h after first CR feeding). Gray arrow represents colostrum meals fed within 1 h after birth (8% of BW) and 12 h after the first CR feeding (4% of BW) for calves assigned to LF. Black arrows represent colostrum meals fed within 1 h after birth (4% of BW) and 6 (4% of BW) and 12 h after the first CR feeding (4% of BW) for calves assigned to HF. Asterisks indicate differences between treatments at a given time ($P \leq 0.01$).

be considered that we did not have a treatment where calves were fed only one meal at birth, which limits some comparisons with other studies. Saldana et al. (2019) showed that calves fed a single meal at birth using a lower total IgG mass [IgG concentration of 52.3 g/L (198.74 g of IgG fed)] from maternal colostrum had higher AEA (37.3%) than calves fed colostrum with an IgG concentration of 98.1 g/L (372.78 g of IgG fed; AEA: 20.8%). Even though all calves in the present study were fed colostrum to a total of 12% of their BW in their first 12 h of life, as recommended by Godden et al. (2019), our results indicate that calves fed a decreased total IgG mass in the first meal (4% of BW instead of 8% of BW) did not have an increased AEA. These results disagree with findings from Saldana et al. (2019) and Lopez et al. (2020), who reported decreased AEA when higher doses of IgG were fed. However, it has to be considered that total IgG masses fed by Saldana et al. (2019) in a single meal at 0 h (351.5 g of IgG) were higher than meals in our study. Furthermore, Lopez et al. (2020) also found that calves fed a single meal, with a lower IgG mass of 154 g, within 2 h after birth had an increased AEA (54.4%) compared with calves fed a higher IgG dose of 401 g, which had a lower AEA of 24.4%. In both Saldana et al. (2019) and Lopez et al. (2020), it is important to note that the reported reduction in AEA occurred when calves were fed a higher total IgG mass at birth, delivering a high amount of total IgG (>300g). This might indicate that a higher IgG mass needs to be fed in a single meal to induce a possible upper limit of IgG absorption.

In addition to the studies performed by Saldana et al. (2019) and Lopez et al. (2020), which reported IgG absorption when calves were fed only a single colostrum meal after birth, other studies have also researched colostrum feeding frequencies involving more than one meal (Morin et al., 1997; Jaster, 2005; Cabral et al., 2012, 2014). Morin et al. (1997) evaluated feeding 1 or 2 colostrum meals. Morin et al. (1997) studied the effects of feeding 4 L at birth or 4 L at birth followed by a 2-L meal of maternal colostrum with an IgG concentration of 60.1 g/L at 12 h. The authors reported that feeding 2 meals (4 L at birth plus 2 L at 12 h) resulted in higher serum IgG concentrations. The current study, however, did not compared feeding 2 or 3 meals against one single meal, where possible reductions of IgG absorption have been observed (Morin et al., 1997; Saldana et al., 2019). Similar to the 2 feeding frequencies studied by Morin et al. (1997), Cabral et al. (2012) compared the effects of feeding CR at 0 h (191.4 g of IgG fed) or at 0 and 6 h (127.6 and 63.8 g of IgG fed, respectively) to newborn Holstein calves. Cabral et al. (2012) reported that serum IgG was not different for calves fed once or twice (15.8 and 16.8 g/L, respectively). In addition, they did not find an increase in AEA when calves were fed twice (32.9%) compared with calves fed once (32.1%). Their results contrast the findings of Morin et al. (1997) but are in agreement with our results, where decreasing total IgG fed at birth by increasing number of meals did not affect IgG absorption. Cabral et al. (2012) discussed that the 6-h meal did not significantly contribute to additional IgG absorption due to decreased or depleted pinocytotic activity, which is responsible for macromolecular transport of IgG after the lumen has initial contact with colostrum at 0 h (Stott et al. 1979b). However, our AUC values indicate that calves fed a lower IgG mass at birth, followed by 2 more colostrum meals (6 and 12 h) were able to achieve the same IgG levels as calves fed higher IgG doses at birth. These findings suggest that meals offered at 6 and 12 h for HF calves contributed to additional IgG absorption. In a follow-up study, Cabral et al. (2014) reevaluated the effects of feeding colostrum once versus twice in the first 6 h of life. They compared feeding CR at 0 h (184.5 g of IgG fed) or at 0 and 6 h (123.0 and 61.5 g of IgG fed, respectively). Again, they did not find differences in serum IgG concentrations, AEA, nor AUC at 24 h for calves fed once or twice. In contrast to Cabral et al. (2012) and Cabral et al. (2014), we found that calves fed a higher mass of IgG at birth (LF) had a higher AUC throughout the whole sampling period (6, 12, 24, and 48 h). Nevertheless, even though calves fed LF (colostrum fed at 0 and 12 h) had a consistently greater AUC than HF calves (colostrum fed at 0, 6, and 12 h) up to 24 h, serum IgG values at 24 h were not different. It was not directly measured in this study, but the higher AUC for LF fed calves could imply that the rate of absorption was higher during the first hours of life due to the greater total mass fed compared with HF calves: 257.9 and 128.9 g, respectively. Although AUC was higher for LF calves, serum IgG concentrations at 24 h did not differ. This suggests that HF calves were still able to effectively absorb the IgG fed at 6 and 12 h and still attain the serum IgG concentrations of LF calves, which were fed a higher total IgG mass at birth. From these observations, it could be suggested that calves still absorb IgG efficiently at 6 and 12 h, as they reach sufficient levels of IgG to achieve successful transfer of passive immunity, even when high-quality colostrum feeding is delayed to 6 or 12 h after birth (Fischer et al., 2018). Hare et al. (2020) also found that calves fed prolonged colostrum meals after a first meal at birth (every 12 h from 12 until 72 h) had increased serum IgG concentrations and demonstrated an apparent IgG persistency of 88.2%. From these experiments studying feeding regimens, only Jaster (2005) fed a high dose of IgG at birth (>300 g) and consequently found a response in IgG uptake and AEA. Jaster (2005) evaluated feeding 4 L of high-quality maternal colostrum (84 g/L of IgG) once (0 h; 336 g of IgG fed) or feeding 2 L at 0 h (168 g of IgG fed) and 2 L at 12 h (168 g of IgG fed) to newborn Jersey calves. They reported that serum IgG concentrations at 24 h were higher for calves fed 2 L twice (45.8 g/L) rather than feeding one meal of 4 L (39.7 g/L). In addition, AEA at 48 h was increased when calves were fed 2 separate 2-L meals (31.2%) instead of a single 4-L meal of high-quality colostrum (24.6%). Overall, providing a second colostrum meal to newborn calves seems to contribute to circulating serum IgG concentrations at 24 h.

Jaster (2005) fed a considerably higher total IgG mass (336 g) to calves in a single meal compared with the doses fed by Cabral et al. (2012; 191.4 g), Cabral et al. (2014; 184.5 g), and our study (128.9 g). This leads us to speculate that high doses of IgG are needed to decrease AEA and IgG absorption. Saldana et al. (2019) and Lopez et al. (2020) reported decreased AEA when high total IgG masses, above 300 g, were fed in a single meal after birth. In contrast, the present study only fed calves either 258 or 126.7 g of IgG at birth. Besser et al. (1985) noted that newborns have a physiological limit to the amount of Ig they can absorb from a given volume of colostrum. Moreover, Saldana et al. (2019) suggested that a decrease in AEA is due to an upper limit of absorption in a given period. Although total IgG mass fed is a limiting factor, total volume fed is also an important consideration. Stott and Fellah (1983) reported that calves absorb IgG more efficiently when a determined total IgG mass is fed in 1 L rather than 2 L of colostrum. Specifically, Stott and Fellah (1983) stated that 1 L of colostrum with an IgG concentration of 100 g/L is better absorbed than 2 L of colostrum with an IgG concentration of 50 g/L. Moreover, Jaster (2005) also discussed how large volumes of colostrum are absorbed less efficiently.

Colostrum replacer AEA varies between different products due to different manufacturing techniques (Godden et al., 2009; Lopez et al., 2020). The AEA from maternal colostrum can range from 20 to 35% (Jones and Heinrichs, 2006); however, these ranges vary, with Halleran et al. (2017) reporting a range from 7.7 to 59.9% in dairy heifers fed a standardized meal, whereas Quigley et al. (2019) and Lopez et al. (2020) reported AEA ranges from 15.0 up to 40.1% for calves fed CR products. The reasons why AEA varies between calves remain unclear, but colostrum composition can be a factor. For example, the removal of casein and fat in CR resulted in AEA values of 40 and 41% (Lopez et al., 2020), whereas high doses (>500 g) of casein added to colostrum affected colostral Ig absorption (Davenport et al., 2000). Finally, other factors, such as maternal colostrum pasteurization and the presence of bioactive hormones (e.g., IGF-1), have been shown to positively and negatively affect AEA (Gelsinger et al., 2014, 2015; Hammon et al., 2020; Pyo et al., 2020). Although CR composition is variable, in this study we used dried bovine colostrum derived from whole colostrum; thus we anticipate our results would be similar to calves fed maternal colostrum.

Although we did not find any benefit to feeding 3 meals of colostrum, none of the calves in either of the 2 feeding regimens in this study had failed transfer of passive immunity or were categorized as having poor (<10 g/L) or fair (10.0 to 17.9 g/gL) IgG levels, as proposed by Lombard et al. (2020). This demonstrates that neither of these feeding frequencies negatively affected serum IgG levels or compromised transfer of passive immunity. Therefore, increasing colostrum feeding frequencies in the first 12 h of life when feeding volumes according to industry standards, >10% of birth BW, with colostrum containing an IgG concentration of 70.5 g/L, does not result in added benefits, such as increased serum IgG concentrations at 24 h. This underlines that colostrum management does not require dairy famers to offer 3 colostrum meals in the first 12 h of life.

CONCLUSIONS

These results indicate that additional meals of CR with a 70.5 g/L IgG concentration do not result in added benefits to serum IgG concentrations or AEA levels. Feeding newborn calves a total IgG mass of 257.98 g does not appear to influence a possible upper limit of IgG absorption or decrease AEA; thus, a higher IgG mass fed at birth is needed to negatively affect AEA or IgG absorption. These results indicate that feeding additional meals of high-quality colostrum do not result in added benefits to serum IgG or AEA levels.

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